

CO₂-AT₃
Class Test

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Sub Code: ITA0502

Sub Name: Computer Vision

① Image Enhancement in Spatial Domain

Defination: Image enhancement is the Process of improving the visual quality of an image by modifying its Pixel values directly.

$$g(x, y) = T[f(x, y)]$$

Techniques:-

i) Gray Level Transformation:-

- changes Pixel intensity values
- Examples:- Negative, Log Transformation, Gamma correction

ii) Histogram Processing:-

- Improves image contrast by redistributing Pixel intensities
- Histogram equalization is the most common method.

iii) Spatial Filtering:-

- Uses a mask to Process neighbouring Pixels
- Smoothing filters remove noise.

Applications:-

- medical imaging
- Face recognition.
- Industrial inspection.

② Image Smoothing

- Removes noise from an image
- Uses Low-Pass Filters.
- Produces a blurred image
- Examples:- Mean, median and Gaussian filters

Image Sharpening

- Enhances edges and fine details.
- Uses High-Pass Filters.
- Produces a sharper image.
- Examples:- Laplacian, Sobel and unsharp masking

Difference

Smoothing

- Removes noise
- Blurs image
- Low-Pass Filter

Sharpening

- Enhances edges
- Sharpens image
- High-Pass Filter

Applications:-

- Smoothing:- medical images noise reduction.
- Sharpening:- object detection, OCR, face recognition.

③ Frequency Domain

Defination:- Frequency domain enhancement improves

image quality by modifying the frequency components of an image using the Fourier transform.

Process:-

- convert image into frequency domain
- Apply frequency filter
- convert back using inverse Fourier Transform.

Types of filters:-

Low-Pass Filter:-

- Removes high-frequency noise
- Produces smooth images

High-Pass Filter:-

- Enhances high-frequency components.
- Improves edges and images sharpening

Applications

- Noise removal
- Medical imaging
- satellite image Processing
- Edge Enhancement

4) Image Segmentation Techniques Comparison

⇒ Segmentation Partitions an image into distinct, non-overlapping regions based on features like intensity, color, or texture to isolate objects from the background.

Comparison matrix:-

- Point Detection:-
 - uses Laplacian mask to find isolated intensity spikes
 - Fast, very simple
 - sensitive to noise
 - Semiconductor surface defect detection
- Line Detection:-
 - Directional masks ($0^\circ, 45^\circ, 90^\circ, 135^\circ$)
 - Directs orientation
 - Needs fixed line width
 - PCB trace checking.
- Edge Detection:-
 - Derivative operators (Sobel, Canny)
 - Captures boundaries
 - Fragmented outputs.
 - Shape detection setup
- Thresholding:-
 - Binary cut via intensity $T: f(x,y) > T$.
 - Minimal compute
 - Fails under non-uniform light.
 - Text binarization
- Region - Based:-
 - Groups similar adjacent pixels (growing/splitting).
 - ~~to~~ closed boundaries
 - High compute cost.
 - medical tumor scanning

Edge Detection, Linking and Boundary Detection

3-step Pipeline

i) Edge Detection:- Locates gradient discontinuities using Sobel or Canny (Gaussian smoothing $\rightarrow$ Gradient calculation $\rightarrow$ Non-maximum suppression $\rightarrow$ Hysteresis thresholding).

ii) Edge Linking:- connects broken ~~edges~~ edges segments using.

$\Rightarrow$ Local Processing:- Groups Pixels matching Local gradient magnitude and direction tolerance (E_m, E_a).

$\Rightarrow$ Hough Transform:- maps points to Parameter space ($p = x \cos \theta + y \sin \theta$) to Link collinear Lines.

iii) Boundary Detection:- Uses contour-following or Active Contours (snakes) to extract closed boundaries for object classification and quality inspection.